ADVANCED UNIT

Advanced Water Safety

A six-lesson unit on the data, physiology, social determinants and prevention strategies behind Victorian drowning — and how to argue for change.

UNIT MAP

Six lessons, one question

VC2HP10P02 · VC2HP10P08

Why does Victoria still drown — despite knowing how to prevent it? Each lesson sharpens one part of the answer.

LESSON 1

01

Advanced data analysis — Victorian & global drowning trends

VC2HP10P02 · VC2HP10P08

LESSON 2

02

Physiology of drowning & CPR mechanics

VC2HP10P08 · VC2HP10M01

LESSON 3

03

Social determinants of drowning risk

VC2HP10P02 · VC2HP10P08

LESSON 4

04

Prevention strategies & international comparison

VC2HP10P08 · VC2HP10P09

LESSON 5

05

Diving, spinal injuries & high-risk teenage behaviour

VC2HP10P02 · VC2HP10P08

LESSON 6 · ASSESSED

06

Assessment task — planning, drafting & peer review

VC2HP10P08 · VC2HP10P09

PREREQUISITE · BUILT ON YEAR 7–8

You already know the rules. Now we ask why they aren't enough.

YEAR 7–8 · YOU CAN ALREADY

- Read the six Australian beach flags
- Recall the five rip-current survival steps
- Execute DRS ABCD in order
- Apply the five LSV SwimSafe rules
- State the headline Victorian drowning numbers

YEAR 9–10 · WE NOW ASK

How do you **calculate** a drowning rate and read a trend?

What **physiologically** happens in the four minutes after submersion?

Which **structural factors** decide who drowns?

Which prevention strategies are **backed by evidence** — and which aren't?

What would **you** recommend to a Minister?

BRIDGE

IF YOU HAVEN'T DONE Y7–8 · ASK FOR THE CATCH-UP SUMMARY

LESSON 1 · 50 MIN · WORKBOOK

01

Advanced data analysis — Victorian & global drowning trends

VC2HP10P02 VC2HP10P08 CLASSROOM · INDIVIDUAL + PAIRS

THE UNIT'S BIG QUESTION — RETURN TO IT IN LESSON 6

“

Why does Victoria still drown — despite decades of awareness campaigns, training, and warning signs — and what would actually move the numbers?

L1 · OPEN

SOURCE · LSV VICTORIAN DROWNING REPOR

The numbers behind one Victorian year

FATAL DROWNINGS

52

Lives lost in Victorian waters, 9% above the ten-year average of 48.

NON-FATAL DROWNINGS

+37%

123 non-fatal drownings, above the ten-year average. The near-misses show more strain than the death toll alone.

MULTICULTURAL COMMUNITIES

37%

Of deaths since 2015–16. Country of birth was unknown in a further 22%, so this is a lower bound.

IN PUBLIC POOLS

0

Zero of the 52 deaths occurred in a public swimming pool — the first time since 2019–20.

SOURCE · LSV.COM.AU/RESEARCH

37%

of Victorian drowning deaths since 2015–16 were people from multicultural backgrounds — 185 deaths. Country of birth was unknown in a further 22% of cases, so 37% is a lower bound.

This is **over-representation**. Lesson 3 will ask **why** — and what a school can do about it.

SOURCE · LSV VICTORIAN DROWNING REPOR

WORKBOOK ACTIVITY 1.1A · CALCULATE THE RATE

Victoria · Australia · the world — per 100,000 people

VICTORIA · 2024–25

52 ÷ 6,900,000 × 100,000

0.75

per 100,000 people · LSV 2024–25

AUSTRALIA · ANNUAL AVG

RLSSA national data

1.2

per 100,000 people · RLSSA

GLOBAL · HIGH-INCOME

WHO benchmark

~0.8

per 100,000 people · WHO

In your workbook: show your working. If Victoria matched the 0.8 high-income benchmark, that's about **55 → 55** deaths — but the 60% jump in non-fatal incidents is where the real warning sits. (See Activity 1.1c.)

SOURCES · LSV · RLSSA · WHO.INT

GLOBAL CONTEXT · WHO 2021

Drowning is one of the world's most unequal causes of death

GLOBAL DEATHS / YEAR

~320k

Around 300,000 people drown each year worldwide; 92% in low- and middle-income countries (WHO, 2024).

IN LOW & MIDDLE-INCOME

90%

Of global drowning deaths happen in low- and middle-income countries.

GLOBAL · MALE VICTIMS

~80%

Of global drowning deaths are male — a pattern repeated in Victorian data.

HIGHEST-RISK AGE

<5

Children under five are the highest-risk age group globally.

SOURCES · WHO.INT · RLSSA · CDC

In one sentence — what is the single most important data insight from today, and why does it matter?

Workbook open to Section 1. Notes closed. **Insight** — not just a number. The most powerful answers will name a number **and** what it implies.

LESSON 2 · 50 MIN · WORKBOOK

02

Physiology of drowning & CPR mechanics

VC2HP10P08 VC2HP10M01 INDIVIDUAL

Five stages — from submersion to brain damage

01

Submersion

Face goes under. Voluntary breath-holding begins. The clock starts.

02

Laryngospasm

Involuntary airway closure. In ~10–15% of cases water is kept *out* — "dry drowning" — but asphyxia continues.

03

Hypoxia

Oxygen levels drop at the tissue level. The casualty loses consciousness; reflexes fade.

04

Cardiac arrest

Without oxygen the heart fails to pump effectively. Circulation stops; bystander CPR is now the only intervention.

05

Cerebral hypoxia

Brain cells start dying within 4–6 minutes. After 10 minutes without CPR, survival without major damage is unlikely.

SOURCES · ST JOHN VIC · HEART FOUNDATION · WHO

MINUTES

4

The window in which bystander CPR makes the biggest difference between full recovery and lasting brain damage.

Chest compressions deliver roughly **25–30%** of normal cardiac output — enough oxygenated blood to keep the brain alive until the heart can restart.

Starting at **4 minutes** produces dramatically better outcomes than starting at **8**.

SOURCES · HEARTFOUNDATION.ORG.AU ·

What chest compressions actually do

30 COMPRESSIONS + **2** BREATHS

WHY IT WORKS

Compressions create ~ **25–30% of normal cardiac output** . That's enough oxygenated blood to keep the brain viable until the heart restarts — spontaneously, or with a defibrillator.

Rate: **100–120 beats per minute** . Don't stop until paramedics take over or the casualty is breathing.

SOURCE · ST JOHN AMBULANCE VICTORIA

THREE PATTERNS EVERY RESPONDER SHOULD KNOW

Wet · dry · secondary — and the 1-10-1 cold-water rule

A · WET DROWNING

Water enters the lungs

The most common pattern. Water in the lungs disrupts gas exchange and triggers cardiac arrest.

B · DRY DROWNING

Laryngospasm seals the airway

In ~10–15% of cases, water never reaches the lungs — but the casualty still dies of asphyxia.

C · SECONDARY DROWNING

Symptoms hours later

Always seek medical review after any submersion — coughing, fatigue or chest pain can appear hours after the event.

COLD WATER · 1-10-1

Three survival windows

`1 MIN` control breathing · `10 MIN` useful swimming · `1 HR` before hypothermia incapacitates.

SOURCES · ST JOHN VIC · RLSSA

Explain the 4-minute window to someone who has never heard of it.

Plain language. **No medical jargon.** If a Year 5 student can repeat your explanation back to you, you've nailed it.

LESSON 3 · 50 MIN · WORKBOOK

03

Social determinants of drowning risk

VC2HP10P02 VC2HP10P08 PAIRS · JIGSAW

FRAMING THE QUESTION

"They made a bad choice" isn't the whole story

INDIVIDUAL EXPLANATION

Personal behaviour caused the outcome.

"He shouldn't have been swimming after drinking." "They should have read the sign." "Why did she dive in there?"

True — but incomplete. It can't explain why 37% of deaths since 2015–16 were people from multicultural backgrounds, or a 73% male skew.

STRUCTURAL EXPLANA

Systems shaped who was there, and how.

Cost of lessons. Patrol coverage. Cultural water-fluency. Masculinity norms. Where the unpatrolled rivers are. Who has a backyard pool.

Schools can change **some** of these. That's where prevention starts.

SOURCE · WHO SOCIAL DETERMINANTS OF HEALTH

WORKBOOK ACTIVITY 3.1 · JIGSAW

Six factors that shape who drowns

01 · GENDER NORMS

"Don't back down from a dare." Masculinity expectations are a documented risk factor for young male drowning — and explain part of the 80% male skew.

02 · CULTURAL BACKGROUND

Victorians from multicultural backgrounds may have had fewer opportunities to learn to swim — for reasons of cost, culture, or country-of-origin practice. The drowning data reflects that. LSV reports 58% of fishing-related drowning deaths since 2015–16 were people from multicultural backgrounds.

03 · SOCIOECONOMIC STATUS

Lower-income children are less likely to have learned to swim. Swim lessons aren't free; nor is travel to the coast.

04 · GEOGRAPHY

Rural and regional Victorians swim at unpatrolled inland waterways at higher rates than urban residents — because that's the water they have.

05 · ALCOHOL EXPOSURE

Alcohol is a factor in ~20–30% of adult drowning deaths in Victoria. The social settings in which drinking-and-water mix are themselves structural.

06 · COMMUNITY CAPACITY

Local clubs, multilingual programs and trained bystanders aren't distributed evenly. Where they're missing, drowning is more common.

SOURCE · LSV · RLSSA · WHO

WORKBOOK ACTIVITY 3.1B · PAIRS · 12 MIN

What a Victorian secondary school could actually do

37% of Victorian drowning deaths since 2015–16 were people from multicultural backgrounds. Standard curriculum alone is not closing this gap. In pairs · propose three specific actions — beyond the standard swim & water safety lessons — that your school could realistically implement.

EXAMPLE 01

Multilingual beach orientation

A start-of-summer assembly delivered with EAL/CALD parent groups. Languages chosen from school enrolment data.

EXAMPLE 02

Subsidised lessons partnership

Work with LSV / your local aquatic centre on subsidised secondary-aged lessons for students who haven't had access.

EXAMPLE 03

Peer educator program

Train senior students from CALD communities to deliver short safety briefings to their own family and community networks.

Name ONE social determinant a school can address. ONE it cannot.

Then write a sentence on **why that distinction matters** when you're planning a real prevention strategy.

LESSON 4 · 50 MIN · WORKBOOK SE

04

Prevention strategies & international comparison

VC2HP10P08 VC2HP10P09 INDIVIDUAL + RESEARCH

83%

Reduction in child drowning in home pools attributable to four-sided isolation pool fencing.

The strongest evidence base of any single drowning-prevention intervention. **Engineering > education** when the engineering is right.

Hold this number when we compare strategies on the next slides.

SOURCE · WHO DROWNING PREVENTIO

Six categories of prevention — not all created equal

A · ENGINEERING

Pool fencing, design

4-sided isolation fencing reduces home-pool drowning by ~83%. Strongest evidence of any single intervention.

B · SUPERVISION

Lifeguards & patrols

Zero of the 52 deaths in 2024–25 were in public swimming pools — but 25% were in home environments. Supervision works where it exists.

C · EDUCATION

Swim lessons

Strong evidence for child drowning. For adults, education works best *combined* with behaviour change.

D · LEGISLATION

Laws & standards

Pool-fencing laws, lifejacket requirements, alcohol restrictions near water. Effective when enforced.

E · TECHNOLOGY

AEDs, detection, CCTV

Defibrillators at aquatic facilities, drowning-detection systems, throw-rescue equipment at public sites.

F · COMMUNITY

Programs & campaigns

SwimSafe, Swim & Survive, targeted CALD programs. Strength depends on uptake within high-risk groups.

SOURCES · WHO · LSV TOOLKIT · RLSSA

Five strategies, ranked against the evidence

STRATEGY	EVIDENCE	BEST TARGET GROUP	KEY LIMITATION
Pool fencing (4-sided)	**STRONG**	Children under 5 in home pools	Only addresses backyard-pool drowning
Mandatory primary-school swim lessons	**STRONG**	Children gaining first swim competence	Doesn't reach those who miss school programs
Extending patrol coverage to 7 days	**MODERATE**	Beach-goers in patrolled zones	Doesn't reach inland waterways at all
Warning signs at unpatrolled beaches	**LIMITED**	First-time visitors who read signs	Easily ignored; ineffective with alcohol or peer pressure
Targeted CALD community programs	**MODERATE → STRONG**	The over-represented group	Requires sustained funding & trusted partners

> **Your call · $5m budget:** which two strategies would you fund? (Workbook 4.1a — defend with evidence, not intuition.)

WORKBOOK ACTIVITY 5.1 · COMPARE

Australia, the UK, the US — what each says first

AUSTRALIA · RLSSA

"Swim between the flags."

Patrolled-location centric. Beach-flag literacy, SwimSafe rules, Swim & Survive. Inland waterways named as highest-risk environment.

royallifesaving.com.au

UNITED KINGDOM · RLSS UK

"Float to Live."

Cold-water-shock centric. The primary survival message is a single physiological action — lean back, float, conserve until cold-shock passes.

rlss.org.uk/education

UNITED STATES · RED CROSS

"Water competency."

Skill-centric framework — swimming, breath control, rescue, and lifejacket use packaged as a competency, not a slogan.

redcross.org · take-a-class

In your workbook: identify ONE international element *not* currently prominent in Victorian education. Argue whether Victoria should adopt it — citing one primary source URL.

Which ONE prevention strategy has the strongest evidence? Which has the weakest?

Defend both. One sentence each. **"It feels right"** isn't evidence.

LESSON 5 · 50 MIN · WORKBOOK

05

Diving, spinal injuries & high-risk teenage behaviour

VC2HP10P02 VC2HP10P08

INDIVIDUAL + DISCUSSION

AIHW · RLSSA · ROYAL PERTH SPINAL STUDY

Six numbers that should appear in every prevention message

UNSAFE DIVING — SHARE OF

10%

Of all Australian spinal cord injuries.

UNSAFE DIVING — CAUSE OF

20%

Of quadriplegia cases in Australia.

AQUATIC SPINAL PATIENTS · MALE

95%

A near-total gender skew. **Structural — not coincidental.**

AQUATIC SPINAL · UNDER 30

75%+

Are aged under 30. Highest single age band: **10–14**.

PERMANENT DISABILITY

66%

Of aquatic spinal patients live with permanent disability after the injury.

DROWNING 15–20 YO · POST-COVID

+34%

Increase among 15–20 year olds vs pre-COVID baseline (RLSSA).

L5 · DATA

SOURCES · AIHW · RLSSA · RPH SPINAL STUDY

WORKBOOK ACTIVITY 6.1 · CASE STUDIES

Three locations. One pattern.

CASE 01 · NSW 2025

Terrigal Wharf

A teenager dives from a wharf into water he believed was deep. Helicopter evacuation. **Hidden shallow zone** under structure.

"IT LOOKED DEEP ENOUGH"

CASE 02 · VICTORIA 2025

Torquay Front Beach

Teen suffers a suspected spinal injury after a dive at a Victorian surf beach. Airlifted. **Sandbanks shift with every tide.**

SURF BEACHES ARE NOT "SAFE" FOR DIVING

CASE 03 · NSW 2024

Bawley Beach jetty

A young man is injured diving from a jetty he had used safely before. **New sandbank had formed** beneath the jump zone.

"I'VE DONE IT BEFORE" · FALSE CONFIDENCE

IN PAIRS · 8 MIN · Two factors common to all three. Mechanism — not just behaviour. (Workbook 6.1a.)

SOURCES · ROYAL LIFE SAVING AUSTRALIA · AIHW

Five factors that compound on rivers, dams and lakes

01

Depth changes constantly

Drought, flooding, irrigation releases, sediment movement — even morning to afternoon.

02

Visibility is poor

Submerged logs, branches, debris, even shopping trolleys — invisible under turbid water.

03

Cold-water shock

Rivers and alpine lakes stay cold even in hot weather. The 1-10-1 rule applies.

04

Alcohol + group dynamics

Inland incidents disproportionately involve alcohol, teenage males, and peer-pressured jumps.

05

Remote rescue delay

Limited phone signal, no lifeguards, no AED, slower spinal immobilisation — all worsen outcome.

SOURCE · RLSSA NATIONAL DROWNING REPORT

Six things teenagers believe about diving that are wrong

MYTH

"If others are jumping there, it must be safe."

FACT · depth changes daily. Yesterday's safe jump can be today's spinal injury.

MYTH

"Clear water means deep water."

FACT · clarity is independent of depth. Dangerously shallow water can be crystal clear.

MYTH

"I checked it before — I know this spot."

FACT · Bawley Beach. Familiarity becomes overconfidence becomes injury.

MYTH

"Good swimmers are safe from diving injuries."

FACT · spinal injury happens on impact. Swimming ability is irrelevant.

MYTH

"Feet-first is just as risky as diving."

FACT · feet-first dramatically reduces cervical spine risk. "Feet first, first time."

MYTH

"The jetty is high — so the water must be deep."

FACT · structure height tells you nothing about water depth beneath it.

Feet first, first time.

Never dive into unknown water. Check depth **every single time** , not just the first time. Previous safety at a location does not guarantee current safety.

The highest diving-injury incidence is in 10–14 year olds. Explain why — using social *and* environmental factors.

Then sketch **what a targeted prevention response** would look like. Two sentences each.

LESSON 6 · 50 MIN · WORKBOOK SECTION

06

Assessment task — planning, drafting & peer review

VC2HP10P08 VC2HP10P09 INDIVIDUAL + PEER REVIEW

ASSESSMENT TASK · CHOOSE ONE · BUILD IT FOR A REAL AUDIENCE

Three options — all evidence-based, all argued

OPTION A

Policy Brief

600–800 words to the Victorian Minister for Sport & Recreation. Three recommendations + evidence + measurable outcomes.

Best for: writers who want to argue policy

OPTION B

Communication Campaign

Target ONE high-risk group from Victorian data. Campaign name, message, visual concept, platform + 100-word justification.

Best for: designers, video-makers, social

OPTION C · ADVANCED ★

School Audit & Report

Audit your school's water-safety provision. Benchmark against Curriculum 2.0. Priority matrix + recommendations.

Best for: students who want real impact

All options require: 3+ credible sources with URLs · 1+ Victorian or global statistic · a named target audience · a specific call to action.

Six things that separate "Building" from "Extending"

01 **Beyond description — to analysis**

Argues with evidence. Acknowledges counter-arguments. Doesn't just describe what is.

02 **Primary sources, cited with URLs**

LSV, RLSSA, WHO, AIHW — not random blog posts. The reader can verify every claim.

03 **Structural — not just individual**

Names a mechanism. "Masculinity norms" beats "boys take more risks." Distinguishes systemic from personal.

04 **Built for a named audience**

A Minister's brief doesn't read like a TikTok script. The language fits the addressee.

05 **Measurable outcomes**

SMART metrics. "Reduce CALD drowning deaths by 20% in 5 years" beats "raise awareness."

06 **Your voice is visible**

Not a stitched-together summary. Your teacher should feel *you* wrote this and made the calls.

Six primary sources for the assessment task

Life Saving Victoria

lsv.com.au · lsv.com.au/research

Use for · Victorian drowning data, CALD over-representation, SwimSafe rules, patrolled beach locations.

World Health Organization

who.int/drowning

Use for · Global drowning data, prevention guide, social determinants framework.

International comparators

rlss.org.uk/education · redcross.org/water-safety

Use for · Float to Live, water competency, international approach comparison.

Royal Life Saving Australia

royallifesaving.com.au

Use for · National data, inland waterway research, post-COVID trends, "Feet first, first time."

AIHW

aihw.gov.au

Use for · Spinal cord injury data, quadriplegia statistics, Australian injury surveillance.

Victorian Curriculum 2.0

victoriancurriculum.vcaa.vic.edu.au

Use for · Option C audit · benchmarking school provision against curriculum codes.

Six lessons. One thing each.

LESSON 01

The data

- 52 Victorian deaths · 2024–25
- Non-fatals +37% on 10-yr avg
- 37% multicultural over-representation
- 0 deaths in public pools

LESSON 02

The body

- 5-stage drowning sequence
- 4-minute brain window
- CPR ≈ 30% cardiac output
- 1-10-1 cold-water rule

LESSON 03

The system

- Individual vs structural causes
- Six social determinants

CALD action a school *can* take

- ~80% male skew is structural

LESSON 04

The evidence

- 4-sided fencing −83% child drowning
- Six categories of prevention
- Five strategies, evaluated
- AU vs UK vs US — three messages

LESSON 05

The diving question

- 10% / 20% / 95% / 66%
- Terrigal · Torquay · Bawley
- Inland = 5 compounding factors
- Feet first, first time

LESSON 06

Your argument

- Brief · campaign · audit
- 3+ sources with URLs
- Named audience & SMART metric
- Six rubric criteria · E → A

EMILIE HARRISON · SWIMWATER.SAFETY@GMAIL.COM

Licence & Attribution

© 2026 Emilie Harrison — Swim & Water Safety Teaching Hub

Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). You are free to copy, adapt, remix and redistribute this deck — including commercially — provided you credit the author and indicate any changes.

Attribution: "Swim & Water Safety Teaching Hub" by Emilie Harrison, CC BY 4.0. Source: emilie-harrison.github.io/swim-water-safety

Third-party material is not covered by this licence. Content owned by Life Saving Victoria, Royal Life Saving Australia, Surf Life Saving Australia, VCAA, ACARA, AIATSIS and the Australian Resuscitation Council — including drowning data, clinical guidelines, curriculum content, logos and program names — remains with its owners.

Statistics: Life Saving Victoria, Victorian Drowning Report 2024–25. Deck updated September 2026.

Data sources cited in this deck — retain these citations if you adapt it:

- Life Saving Victoria. (2025). Victorian Drowning Report 2024–25. Life Saving Victoria: Melbourne.
- Royal Life Saving Australia (2026). National Drowning Report 2026. RLSSA: Sydney.
- Surf Life Saving Australia (2026). Summer Coastal Drowning Report 2025/26. SLSA: Sydney.

These figures remain the property of the publishing organisations.